

Behavioral Institutionalism of the Agent-to-Agent (A2A) Economy: The A2A Institutional Dynamics Standard (A2A-IDS)

Dr. V. Dyachkov

Computational Institutional Economics & Multi-Agent Systems Laboratory

Correspondence: itinai.com@gmail.com · SSRN Author ID: 6742298

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ABSTRACT

The Agent-to-Agent (A2A) protocol specification v1.0.1 (Google, 2025; Linux Foundation) establishes syntactic interoperability across heterogeneous software agents via standardized discovery manifests (`/.well-known/agent-card.json`), JSON-RPC 2.0 message framing (`message/send`, `message/stream`), and a deterministic Task lifecycle (`SUBMITTED` → `WORKING` → `COMPLETED` / `FAILED` / `INPUT_REQUIRED` / `CANCELED`). Concurrently, the Model Context Protocol (MCP; Anthropic, 2024) standardizes agent perception filters through structured tool, prompt, and resource binding. However, syntactic conformance provides no instrument for assessing multi-agent institutional resilience, service-level veracity, or cascading fragility. This study formulates and substantiates the *A2A Institutional Dynamics Standard (A2A-IDS)*. We project seven behavioral primitives onto protocol telemetry, formalize the A2A hub as an active informational field characterized by four physical parameters, and define *Institutional Anomie* (A_{hub}) as the Total Variation Distance between declared AgentCard capabilities and empirical log-derived execution clusters. We demonstrate that sustained anomie ($A_{\text{hub}} > 0.30$) triggers cascading orchestration failures, validate econometric stationarity testing (ADF/KPSS) to distinguish true digital institutions from stochastic drifts, and apply Markov-chain spectral gap analysis ($\tau_{\text{rel}} = \frac{1}{1-|\lambda_2|}$) to quantify convergence boundaries. The standard establishes automated governance response rules and backwards-compatible JSON Schema extensions to secure federated agent economies.

Keywords: Agent-to-Agent (A2A) Protocol, Model Context Protocol (MCP), Digital Institutional Economics, Institutional Anomie, Total Variation Distance, Markov Transition Matrix, Spectral Gap, Requisite Variety.

JEL Classifications: C73, D02, D82, L86, O33.

1. INTRODUCTION

1.1. Relevance and Problem Statement

The Agent-to-Agent (A2A) protocol specification v1.0.1 (Google, 2025; Linux Foundation, 2025) establishes syntactic interoperability among heterogeneous software agents through standardized discovery manifests (`/.well-known/agent-card.json`), JSON-RPC 2.0 message framing (`message/send`, `message/stream`), and a deterministic Task lifecycle (`SUBMITTED` → `WORKING` → `COMPLETED` / `FAILED` / `INPUT_REQUIRED` /

CANCELED). However, this syntactic layer provides no instrument for assessing the evolutionary resilience of multi-agent environments. Three structural deficiencies motivate the present study.

First, the specification treats agents as opaque execution units. A hub can observe declared interfaces but cannot audit internal model weights, system prompts, or routing algorithms. Consequently, conformance to protocol grammar does not entail conformance to declared service-level agreements (SLAs), generating an unmeasured gap between normative declarations and executed behavior.

Second, without a telemetry-based measurement apparatus, A2A hubs accumulate latent anomalies—*capability cloaking* (inflated `AgentSkill` declarations in the AgentCard), *cyclic task delegation* (Cycle Overflow across unmonitored agent chains), and *parasitic state retention* (Task Flooding via persistent `INPUT_REQUIRED` loops)—that degrade orchestration quality below detectable thresholds until cascading failure occurs.

Third, the Model Context Protocol (MCP; Anthropic, 2024) and emerging micropayment standards (HTTP 402 / x402) introduce additional interaction surfaces whose institutional stability remains unquantified in current literature on multi-agent systems (MAS), distributed AI governance, and digital institutional economics.

These deficiencies position the A2A hub not merely as a message router but as an institutional environment whose health requires continuous, mathematically grounded diagnostics—a gap this paper addresses.

1.2. Conceptual–Categorical Apparatus

The following constructs form the theoretical vocabulary of the study. Each is operationalized against observable A2A/MCP telemetry rather than inferred mental states:

- **(1) Informational Signal (S):** Any objectively logged event in the hub carrying source, carrier, form, and timestamp fields. In the A2A stack: AgentCard publications, `message/send` payloads, Task state transitions, and `x402` payment headers.
- **(2) Algorithmic Perception Filter (F):** The deterministic or stochastic configuration of a receiving agent that decodes S into an internal context: system prompt, context-window capacity, attached MCP tool set (`tools/list`), token budget, authentication schemas (`securitySchemes` per RFC 7515 / 8785), and LLM weight parameters.
- **(3) Stimulus (σ):** The structured token context assembled in the agent's prompt memory after S passes through F . σ is fully determined by $F(S)$ and contains no anthropomorphic affective component.
- **(4) Reaction (R):** An atomic, logged action or inaction: a Task state transition, emission of a content part (`TextPart`, `DataPart`, `FilePart`), invocation of an external MCP tool via `tools/call`, or connection timeout.
- **(5) Behavioral Pattern (Π):** A reproducible, autocorrelated sequence of reactions exhibited by one agent or a cohort under recurrent signal classes ($r > 0.70$ at lag 1, verified via Ljung-Box test, $p < 0.01$).
- **(6) Stationarity (Σ):** The statistical property that the distribution of reactions preserves its first two moments over successive observation windows W_t , verified via Augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests with $\text{TVD}(P_t, P_{t+1}) < 0.05$.
- **(7) Behavioral Spectrum (K_{spec}):** The full set of stable behavioral clusters ($K \geq 3$) emerging in a heterogeneous agent population in response to a single task class, consistent with Ashby's Law of Requisite Variety (1956).
- **(8) Institutional Anomie (A_{hub}):** The Total Variation Distance between the normative distribution Y (declared in AgentCards and protocol SLAs) and empirical behavioral distribution X (extracted from task logs via unsupervised clustering):

$$A_{\text{hub}} = \text{TVD}(Y, X) = \frac{1}{2} \sum_{i=1}^K |y_i - x_i| \quad (1)$$

- **(9) Agent-as-Proxy:** Every agent is an algorithmic executor of a human or organizational principal. The agent's filter encodes the principal's regulatory constraints, budget limits, and risk tolerances.
- **(10) Informational Field Parameters:** Four physically measurable quantities: signal density D_{hub} , propagation speed V_{prop} , access asymmetry A_{access} (Gini coefficient over message throughput), and Shannon entropy H_{hub} .

1.3. Formal Hypotheses

H1 (Anomie–Fragility Linkage): *If $A_{\text{hub}} > 0.30$ sustained over three consecutive observation windows, the probability of a cascading orchestration failure within the subsequent window exceeds 0.70, controlling for task volume.*

H2 (Spectral Compression under Asymmetry): *An increase in access asymmetry ($\text{Gini}_{\text{msg}} \rightarrow 1$) produces a statistically significant reduction in K_{spec} (toward monoculture), verified by a negative Spearman correlation ($\rho < -0.60$, $p < 0.01$) across hub operational logs.*

H3 (Markov Convergence Criterion): *The hub reaches a stationary institutional equilibrium if and only if the second-largest eigenvalue modulus of the $K \times K$ Markov transition matrix satisfies $|\lambda_2| < 0.85$, yielding a relaxation time $\tau_{\text{rel}} = \frac{1}{1-|\lambda_2|}$ bounded above by a policy-defined threshold.*

H4 (Autonomization Persistence): *Once a behavioral pattern achieves $K_{\text{aut}} > 0.80$, removal or modification of the originating hub signal does not dissolve the pattern within two subsequent windows, indicating self-sustaining institutional inertia.*

1.4. Methodology

The study employs a mixed-methods design combining formal econometric time-series modeling with empirical protocol trace analysis:

- *(a) Unsupervised behavioral clustering:* HDBSCAN (Campello et al., 2013) and Gaussian Mixture Models (Reynolds, 2009) extract natural reaction clusters from task-trajectory feature vectors (state sequences, p50/p95/p99 latencies, token consumption, error codes).
- *(b) Distributional divergence metrics:* Total Variation Distance (TVD) and Jensen–Shannon Divergence (Lin, 1991) quantify gaps between normative declarations (Y) and empirical execution (X).
- *(c) Time-series stationarity testing:* ADF (Said & Dickey, 1984) and KPSS (Kwiatkowski et al., 1992) tests confirm or reject unit roots on cluster-share time series $\{p_{i,t}\}$.
- *(d) Markov-chain spectral analysis:* Transition probability matrices $P = [p_{ij}]$ are evaluated for diagonal dominance ($p_{ii} > 0.85$) and spectral gap ($1 - |\lambda_2|$).
- *(e) Controlled perturbation experiments (A/B):* Hub configuration shifts evaluate pattern autonomization (K_{aut}) and filter feedback rates (Φ_{feed}).
- *(f) Informational-field parameterization:* Continuous telemetry stream computation of D_{hub} , V_{prop} , A_{access} , and H_{hub} .

2. PROJECTION OF SEVEN BEHAVIORAL PRIMITIVES ONTO THE A2A HUB ARCHITECTURAL STACK

The theoretical apparatus of behavioral institutionalism (North, 1990; Ostrom, 1990; Hodgson, 2006; Dyachkov, 2026a-g) is operationalized through seven sequential primitives that map exhaustively onto observable protocol fields:

Table 1. Operational Mapping of Behavioral Primitives to A2A / MCP Protocol Types

Primitive	Formal Specification	Protocol Implementation (A2A / MCP)	Server Telemetry Log Field
Signal (S)	Logged event with source, carrier, form, timestamp	<code>/.well-known/agent-card.json</code> ; <code>message/send</code> JSON-RPC; Task state changes; <code>x402</code> headers	<code>a2a.signal.id</code> , <code>event.timestamp_utc</code>
Filter (F)	Decoding configuration $\sigma = F(S)$	System prompt; context quota; attached MCP tools (<code>tools/list</code>); <code>securitySchemes</code> (RFC 7515/8785)	<code>mcp.tools_hash</code> , <code>agent.context_capacity</code>
Stimulus (σ)	Structured prompt context in token memory	Prompt memory context loaded prior to LLM inference	Inferred via prompt token length
Reaction (R)	Atomic logged machine action or timeout	Task transition (<code>WORKING</code> → <code>COMPLETED</code>); <code>DataPart/FilePart</code> emission; <code>tools/call</code> invocation	<code>task.state</code> , <code>mcp.tool_call</code> , <code>latency_ms</code>
Pattern (Π)	Autocorrelated sequence ($\rho > 0.70$)	Compliant execution chain; clarification loop; delegation cascade	<code>trajectory.ngram_hash</code> , <code>autocorr.rho</code>
Stationarity (Σ)	First two moments invariant across W_t	ADF $p < 0.05$ & KPSS $p > 0.05$; $\text{TVD}(P_t, P_{t+1}) < 0.05$	<code>stats.adf_pvalue</code> , <code>stats.tvd_delta</code>
Spectrum (K_{spec})	Cardinality of viable response modes	Distinct HDBSCAN reaction clusters with share $p_i \geq 0.05$	<code>hdbscan.cluster_count</code> , <code>k_spec.cardinality</code>

Source: Authors' formulation based on Google A2A v1.0.1 and Anthropic MCP (2024).

Algorithmic Perception Filter (F): Critically, F is heterogeneous across agents: two agents receiving an identical signal S will produce different internal stimuli if their filters differ in system instructions, attached MCP tools, or model weights. This heterogeneity is the proximate cause of behavioral spectral diversity and requires no anthropomorphic explanation.

Behavioral Spectrum (K_{spec}): Operationalizing Ashby's (1956) Law of Requisite Variety, spectral width is computed as:

$$K_{\text{spec}} = \sum_{i=1}^K \mathbb{I}(p_i \geq \varepsilon), \quad \varepsilon = 0.05 \quad (2)$$

3. THE A2A HUB AS AN INFORMATIONAL FIELD: FOUR MEASURABLE PARAMETERS

The A2A hub is reconceptualized from a passive router into an active informational field characterized by four physically measurable parameters:

Table 2. Mathematical Definition and Diagnostic Role of Informational Field Parameters

Field Parameter	Formula	Underlying Telemetry	Systemic Institutional Consequence
Signal Density (D_{hub})	$D_{\text{hub}} = \frac{1}{ T } \sum_{t \in T} \frac{M_t + \Delta S_t}{\Delta \tau}$	messages_per_task, tasks_per_agent_hour	Context saturation; forces agents into token evasion or silent timeouts.
Propagation Speed (V_{prop})	$V_{\text{prop}} = p50(\text{TTFM}^{-1})$	Time-to-First-Message (ms); speed Gini G_{speed}	<i>Fait accompli</i> advantage: ultra-fast lightweight nodes capture routing graphs.
Access Asymmetry (A_{access})	$\text{Gini}_{\text{msg}} = \frac{\sum_i \sum_j m_i - m_j }{2N \sum_i m_i}$	Outbound message/send share of top 1% nodes	Spectral compression: monopolistic broad-casters eliminate response diversity.
Field Entropy (H_{hub})	$H_{\text{hub}} = - \sum_{i=1}^K p_i \log_2 p_i$	Empirical share of outcome states in window W_t	Excess entropy indicates malformed payloads, cyclic delegation, and instability.

Formulation based on Shannon (1948) and Gini (1912).

4. FOUR COEXISTING INSTITUTIONAL REGIMES AS HYBRID CONFIGURATIONS

Empirical observation of enterprise systems confirms that multi-agent adoption does not proceed as a binary shift from human to autonomous execution. Rather, four interaction regimes operate simultaneously within the A2A infrastructure:

Table 3. Four-Regime Interaction Matrix with Protocol Primitives and Telemetry Standards

Regime	Sender → Receiver	Signal & Filter Architecture	Behavioral Spectrum	Diagnostic Governance Metric
H2H	Human → Human	Formal A2A Task; Filter: role authority, cognitive capacity	Accept, Delegate to Algorithmic Agent, Escalate, Reject, Request Info	human_response_latency, human_escalation_rate
H2A	Human → Agent	Natural intent parsed to Task; Filter: system prompt, MCP tools	Execute, Request Clarification (INPUT_REQUIRED), Policy Block, Sub-delegate	intent_to_task_conversion_rate, clarification_count
A2H	Agent → Human	INPUT_REQUIRED / AUTH_REQUIRED state; Filter: operator attention	Authorize, Ignore/Timeout, Re-route to Expert, Force Cancel (CANCELED)	escalation_response_time, escalation_abandonment_rate
A2A	Agent → Agent	JSON-RPC message/send, ×402; Filter: opaque model weights	Complete (COMPLETED), Critical Fail (FAILED), Re-delegate, Timeout	task_completion_rate, delegation_depth, inter_agent_error_rate

Tripartite Coexistence: Three institutional arrangements persist in stationary equilibrium: (a) *Legacy Manual-Patched Workflows*, resolving algorithmic edge cases; (b) *Hybrid Human-in-the-Loop Automation*, maintaining financial authority and legal accountability (Dietvorst et al., 2015, 2018); and (c) *Fully Autonomous A2A Automation*, confined to high-frequency, low-liability circuits (e.g. ×402 computational token settlements).

5. A2A ANOMIE: THE MEASURABLE GAP BETWEEN DECLARATION AND BEHAVIOR

In the A2A ecosystem, institutional anomie (Durkheim, 1897; Merton, 1938) is operationalized as the divergence between the machine-readable **Normative Layer** (Y) broadcast in AgentCards and the empirical **Behavioral Layer** (X) observed in task traces.

For continuous governance optimization, the smooth Jensen–Shannon Divergence complements TVD:

$$\text{JSD}(Y \parallel X) = \frac{1}{2} D_{\text{KL}}(Y \parallel M) + \frac{1}{2} D_{\text{KL}}(X \parallel M), \quad M = \frac{1}{2}(Y + X) \quad (3)$$

Table 4. Diagnostic Anomie Thresholds and Automated Hub Governance Actions

Diagnostic Zone	A_{hub} Range	Institutional State	Automated Governance Protocol Response
Stability Zone	$A_{\text{hub}} < 0.10$	Equilibrium; declarations reflect execution	Nominal operation; continuous JWS / RFC 8785 signature audit.
Spectral Expansion	$0.10 \leq A_{\text{hub}} \leq 0.30$	Filter drift, schema mismatches, retry spikes	Entropy-reduction: schema validation enforcement, suspect rate-limiting.
Fragility Zone	$A_{\text{hub}} > 0.30$	Severe decoupling; systemic collapse risk	Quarantine anomalous agents, revoke routing rights, mandatory HITL audit.

Worked Numerical Trace: An orchestrator declares $Y = (0.95, 0.05, 0.00)$ for SLA compliance, high latency, and critical failures. Empirical telemetry reveals $X = (0.35, 0.45, 0.20)$. The anomie index computes to:

$$A_{\text{hub}} = \frac{1}{2} \left(|0.95 - 0.35| + |0.05 - 0.45| + |0.00 - 0.20| \right) = \frac{1}{2} (0.60 + 0.40 + 0.20) = 0.60 \quad (4)$$

Because $0.60 \gg 0.30$, the node is quarantined immediately, preventing systemic propagation of routing failures.

6. HUB STABILITY: THE MEASUREMENT PROTOCOL

The measurement protocol evaluates multi-agent stability through four stages:

Stage 1 (Sliding Windows): Telemetry events are binned into observation windows W_t ($\Delta t = 15$ min for high-frequency trading; $\Delta t = 24$ hours for enterprise workflows), forming probability vectors $P_t = (p_{1,t}, \dots, p_{K,t})$.

Stage 2 (TVD Drift Tracking): Inter-window drift is computed as $\text{TVD}(P_t, P_{t+1}) = \frac{1}{2} \sum_i |p_{i,t} - p_{i,t+1}|$. Values below 0.05 indicate local institutional stationarity.

Stage 3 (Econometric Stationarity): Cluster share series $\{p_{i,t}\}$ are submitted to Augmented Dickey–Fuller regressions:

$$\Delta p_{i,t} = \alpha + \beta t + \gamma p_{i,t-1} + \sum_{j=1}^k \delta_j \Delta p_{i,t-j} + \varepsilon_t \quad (5)$$

Rejecting the unit-root null ($p < 0.05$) alongside KPSS failure to reject confirms a stable digital institution.

Stage 4 (Markov Spectral Analysis): Transition probabilities $P = [p_{ij}]$ are evaluated for diagonal retention ($p_{ii} > 0.85$) and relaxation time:

$$\tau_{\text{rel}} \approx \frac{1}{1 - |\lambda_2|} \quad (6)$$

As $|\lambda_2| \rightarrow 1$, relaxation time diverges toward infinity, demonstrating loss of self-regulatory equilibrium and the onset of infinite delegation cycling (Cycle Overflow).

7. MEASURABLE INSTITUTIONAL GAPS OF THE A2A HUB

Economic Vacuum: The base A2A specification provides no native escrow, staking, or reputation accounting. Agents face zero economic penalties for task abandonment. Augmenting the protocol with HTTP 402 / `x402` headers and smart-contract escrow eliminates this moral-hazard vulnerability (Arrow, 1985; Weber et al., 2019).

Opaque Execution Gap: RFC 7515 (JWS) guarantees sender identity, not semantic quality. Formal task completion (COMPLETED) does not guarantee validity. Zero-knowledge inference circuits and verifiable cryptographic execution traces are required to bridge this gap.

Human-in-the-Loop Guarantees: The interrupted states (`INPUT_REQUIRED`, `AUTH_REQUIRED`) represent native institutional circuit breakers that ensure compliance with AI safety frameworks (EU AI Act, 2024; NIST AI RMF, 2023).

8. HUB ARCHITECTURE WITH MEASURABLE STANDARDS

The standard organizes the A2A hub into five functional layers: (1) *Discovery & Identity* (RFC 8785 JCS canonicalization, RFC 7515 JWS); (2) *Authorization & Escrow* (CAAM, `×402` micropayment escrow locks); (3) *Task Orchestration* (extended eight-state lifecycle: `SUBMITTED` → `WORKING` → [`DELEGATED` | `INPUT_REQUIRED` | `AUTH_REQUIRED`] → [`COMPLETED` | `FAILED` | `CANCELED`]); (4) *Human Escalation* (HITL operator interfaces); and (5) *Institutional Telemetry & Audit*.

Table 5. Specification of Five Novel Institutional Telemetry Metrics

Metric Name	Mathematical Formulation	Operational Interpretation	Governance Action Limit
<code>spectrum_width</code>	$K_{\text{spec}} = \sum_{i=1}^K \mathbb{I}(p_i \geq 0.05)$	Viable reaction cluster diversity	$K_{\text{spec}} > 3$ triggers schema tightening
<code>anomie_index</code>	$A_{\text{hub}} = \frac{1}{2} \sum_i y_i - x_i $	Normative vs. empirical SLA divergence	$A_{\text{hub}} > 0.30$ triggers quarantine
<code>stability_tvd</code>	$\text{TVD}(P_t, P_{t+1}) = \frac{1}{2} \sum_i p_{i,t} - p_{i,t+1} $	Inter-window behavioral drift	$\text{TVD} \geq 0.10$ signals restructuring
<code>autonomization_score</code>	$K_{\text{aut}} = \mathcal{E}_{\text{preserved}} / \mathcal{E}_{\text{baseline}} $	Pattern retention following signal perturbation	$K_{\text{aut}} > 0.80$ confirms institutional inertia
<code>filter_feedback_rate</code>	$\Phi_{\text{feed}} = N_{\text{distorted}}/N_{\text{total_signals}}$	Distortion induced by accumulated error history	$\Phi_{\text{feed}} > 0.20$ requires memory reset

9. CONCLUSIONS AND SYNTHESIS OF FINDINGS

This study formulates and substantiates the **A2A Institutional Dynamics Standard (A2A-IDS)**, bridging behavioral institutional economics with the A2A v1.0.1 and MCP protocol stacks. By parameterizing the hub informational field, formalizing institutional anomie (A_{hub}) via Total Variation Distance, and deploying econometric stationarity tests alongside Markov spectral relaxation analysis ($\tau_{\text{rel}} = \frac{1}{1-|\lambda_2|}$), hub operators acquire an objective mathematical instrument stack for detecting structural fragility and preventing catastrophic multi-agent cascade failures before they materialize.

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